



Miguel Peixoto

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EXPERIENCE

- **PeixotoLabs** Portugal
Founder Jun 2026 – Present
AI and engineering company: software, ML, electronics and automation for industrial and operational problems.
- **Sybilion** Porto, Portugal
Founding Machine Learning Engineer Jan 2026 – Oct 2026
 - **Forecasting:** Built the internal forecasting machinery from scratch (8-stage pipeline with HPO) over 3B+ time series, and trained SSL time-series embedding models from scratch (128-d, contrastive and reconstruction).
 - **Agents:** Multi-agent competitive arenas (fan-out, score, select) using orchestrator-worker setups, bounded tools, state machines, and metrics gates.
- **Keenfinity (Bosch Spin-off)** Portugal
AI Architect Feb 2025 – Dec 2025
 - **Leadership:** Leading team of data, frontend and backend engineers across 16 projects in 7 departments.
 - **Projects:** Training LLMs from scratch, deploying GenAI frontends/backends, autoencoders for anomaly detection, PDF information extraction/classification/clustering, automatic ticketing with Agentic LLMs, Computer Vision for factory traceability, on-premises system design on 30+ linux machines for sensitive applications.
- **Bosch Security Systems** Portugal/Germany
Boost Program - Machine Learning Engineer Aug 2023 – Feb 2025
Selected as 1 of 8 from 900+ applicants for Portuguese JMP. Mentored by Sergio Salústio, Director of R&D.
 - **Computer Vision:** Designed and deployed a computer vision and object detection system for real-time factory material tracking, optimizing logistics and reducing inefficiencies. Solution scaled to 3 factories globally.
 - **Embedded ML:** Full ML lifecycle on embedded devices: Data engineering, training, finetuning, distillation, quantization, pruning, compilation and firmware deployment.
 - **AI Solutions:** Engaged stakeholders to identify use cases and deploy solutions while ensuring data governance.
- **University of Minho** Braga, Portugal
Researcher May 2023 – Dec 2024
 - **Industry:** Collaborated with Bosch Car Multimedia to advance anomaly detection in industrial applications.
- **Instrumentation and Experimental Particle Physics Laboratory** Braga, Portugal
Researcher Oct 2021 – Dec 2024
Led team of 10 students specializing in anomaly detection and quantum computing for high-energy physics.
 - **NLP & Healthcare:** Collaborated with SPAC Lab on disease diagnosis model using Portuguese National Health Service records. Implemented HuggingFace Sentence-Transformers and Facebook Faiss for similarity search.
 - **Quantum ML:** Compared Quantum ML architectures with classical methods. Developed robust library with parallelization, unit tests, CI/CD pipelines using PennyLane, Qiskit, PyTorch, and Optuna.
 - **Education:** Facilitated machine learning workshops and seminars for research institute students.
- **Instrumentation and Experimental Particle Physics Laboratory** Braga, Portugal
Summer Intern Apr 2020 – Oct 2021
 - **Research Projects:** Advanced data analysis for dark matter research (2020) and Anomaly Detection for New Physics Discovery at CERN's Large Hadron Collider (2021).

EDUCATION

- **University of Minho** Braga, Portugal
Master's Degree in Information Physics; Grade: 20.0/20.0
 - **Thesis:** Anomaly Detection as a Quality Control Tool in an Industrial Context. Production-ready AutoML autoencoder framework, validated at Bosch Braga on defect-free data only; outperformed SOTA on MvTec. Supervised by Prof. Nuno Castro, Director of LIP and RCNA.
- **University of Minho** Braga, Portugal
Bachelor's Degree in Physics Engineering; Average Score: 16.4/20.0
 - **Leadership:** Founded NEFUM Discord community (200+ members), organized national physics students meeting (134 attendees), ran for association president.
 - **Skills Development:** Member of Academic Debate Association, developing communication and analytical skills.

PUBLICATIONS

- **Automated Anomaly Detection for Manufacturing:** Self-Tuned Autoencoders with Synthetic Defects. TechRxiv preprint, 2025. DOI: 10.36227/techrxiv.176153508.89503711/v1
- **Fitting a Collider in a Quantum Computer:** Tackling the Challenges of Quantum Machine Learning for Big Datasets. Frontiers in Artificial Intelligence, 2023. DOI: 10.3389/frai.2023.1268852

AWARDS & HONORS

- **15th Fraunhofer Portugal Challenge:** Outstanding innovation in technological research for master's thesis.
- **UMinho University Award:** Initiation in Scientific Research award for Variational AutoEncoders anomaly detection system in high-energy physics. [University Media Link] [Local Newspaper]

- **Empreende@Villa.Jovem-BILATECH:** 1st place and 5k€ prize among 30+ finalists in the contest aimed at young entrepreneurs for innovative business idea in marketing automation.
- **Meritorious Behavior Award:** Awarded 4 distinctions at the end of high school due to my participation in "Project Rocket" and my seminar on the Noble Prize of Physics 2017 promoted by UTAD University.

SOCIAL AND POLITICAL ACTIVITIES

- **European Researchers' Night - LIP Presentation** Braga, Portugal
Science Outreach Presenter *2022, 2023*
Presented SOTA physics research to the public.
- **iQuHACK MIT Hackathon** Cambridge, MA, USA
Quantum Computing Participant *2022*
Built solutions on IonQ quantum hardware.
- **ESA Winter Astronomy Camp** Saint-Barthélemy, Italy
Selected Participant *2019*
Selected as 1 of 40 from 500+ worldwide applications; exoplanet research and observatory work.